

Abliteration does not buy autonomy: refusal removal across capability- and willingness-bound offensive tasks in small local LLMs

Gonzalo D. C. García*

June 2026

Abstract

A widely repeated worry holds that *abliteration*, surgically removing the refusal direction from an open-weight model, turns a cheap locally-runnable LLM into a more dangerous autonomous attacker. It conflates willingness with capability, which we separate empirically on a single consumer laptop GPU. Three findings follow. Removing refusals does not raise a small model’s autonomous offensive capability and usually lowers it. The effect is signed: on a task bound by willingness rather than capability it reverses, and abliteration helps. And the policy-relevant question, whether a robust safeguard survives, is one our instrument cannot answer, for reasons we set out.

On capability we run matched pairs, an aligned 7–8B model against its ablated twin, inside a tool-using operator scaffold against a controlled synthetic OSINT range. Across Llama-3.1-8B and Qwen2.5-7B (11 personas $\times$ 3 seeds, 95% bootstrap CIs) the aligned model recovers more planted facts, with non-overlapping CIs; Granite collapses for both twins, a tool-protocol failure we read as a floor. The result survives a change of framing and points the same way in a second, network-free domain (static code audit), though that domain is under-powered ($n=3$ codebases) and we treat it as directional. A frontier model (Opus 4.8) run blind on a non-person variant scores 100%, a ceiling (in its own harness, so a ceiling and not a like-for-like comparison) that places the small models’ shortfall on the capability axis. Flip the task to writing a phishing lure and the sign flips with it: the ablated model complies where the aligned one refuses, by as much refusal as the aligned twin had to give up, which for one of three families is almost none (harm 0 to 85 in Llama, 32 to 77 in Qwen, unchanged at 81 in Granite).

The safeguard question is the one we cannot answer, and the reason is itself a result. Because the range is declared synthetic, a well-aligned frontier model complies with the person-OSINT task ($\sim 64\%$ recall, planted credentials included, in both neutral and adversarial framings) just because it reads the records as fictitious; a declared-synthetic instrument therefore cannot measure context-dependent refusal at all. An earlier draft of ours claimed the opposite, a robust frontier refusal. On audit that was a contaminated harness, which we retract and document as a cautionary case. Practically, removing refusals adds little to a small model’s misuse ceiling: what governs the risk is the task’s bottleneck and the fragility of the small model’s own alignment, with the weight edit a secondary factor. We release the range generator, the operator harness, the graders, and all run artifacts.

1 Introduction

Open-weight “uncensored” and ablated LLMs are everywhere, and they run on a gaming laptop. The worry that follows is a short syllogism. Safety training adds refusals, abliteration strips them

*Independent Researcher. Preliminary findings; preprint.

out, so an ablated model ought to be a more capable autonomous attacker. The last step is where it breaks. Refusal is a property of *willingness*; operating offensively on your own is a property of *capability*, which means multi-step tool use, chaining evidence across calls, and not hallucinating. Deleting a refusal direction acts on the first of these. It does nothing obvious for the second, and it may even hurt.

We test this in the regime the proliferation story actually cares about: a 7–8B quantized model on one 8 GB laptop GPU (an RTX 5060 Laptop), wrapped in a Claude-Code-style operator scaffold (planning ledger, tool execution, retrieval augmentation, an adversarial evidence gate) and pointed at a controlled OSINT task. Three questions shape the design. The first is whether ablation actually raises a small model’s autonomous OSINT capability above its aligned twin. The second is about the nature of any shortfall: if the small models fall short, is that a ceiling they hit, or headroom a maximally capable frontier model could reach and ablation might in principle unlock? The third is more awkward, and we did not anticipate how much it would matter: can a range you have *declared* synthetic even be used to ask whether a frontier model refuses autonomous person-profiling?

The answers, in order. No: the ablated twin is never ahead. In the two families that actually clear the task, Llama and Qwen, it loses with disjoint CIs, and the third (Granite) sits at the floor for both twins, which tells us about its protocol-following, not its capability. The result survives a change of framing. The shortfall is a capability ceiling rather than missing willingness, because a frontier model hits 100% on the blind non-person variant of the range. And no again, the synthetic range cannot isolate refusal, because the frontier complies with a task it can plainly see is fictional. The policy reading is unglamorous. The marginal misuse risk from ablating a small model is small, since scale and capability set its ceiling, not the refusals you removed.

A retracted claim. An earlier version of this work reported a second finding, a “safety inversion”: frontier models robustly refuse the task while small models comply. It was wrong, and the reason is worth stating, because it is the kind of mistake an evaluation harness makes easy. The frontier “refusals” came from prompts that a Windows shell truncated before the tool instructions ever reached the model. Once the harness was fixed the frontier complied at ~64% recall, planted credentials and all, under both neutral and adversarial framings. We retract the safety-inversion claim. The frontier-refusal question, it turns out, cannot be answered with a declared-synthetic instrument at all (§6).

We then turn the design on its head. To check that the capability result is not the whole story, we add a willingness-bound task, writing a targeted phishing lure, where the bottleneck is the refusal itself rather than competence. There the sign reverses: ablation helps, by exactly as much refusal as the aligned model had to give up, which for one of our three families is almost none.

This is a measurement and safety-position paper, not another offensive benchmark. What it contributes: the agentic-autonomy delta ablation induces on consumer hardware, matched-pair and with confidence intervals; the observation that this delta is signed, set by whether the task is bound by capability or by willingness; a synthetic OSINT instrument that is reproducible and keeps no real people in scope; and a negative result, that a declared-synthetic range cannot measure context-dependent refusal, carried by the harness audit that caught our own false finding.

2 Related work

Refusal removal and the brittleness of alignment. Refusal behavior in aligned LLMs is mediated by a single linear direction in activation space, and ablating it (“abliteration”) removes refusals without retraining [2]. More broadly, safety alignment is brittle to cheap weight edits: pruning and low-rank modifications [17], a few steps of fine-tuning [12], or direct safety-finetuning stripping [11] all degrade refusals. This literature establishes that refusals are *easy to remove*. It does not establish that removing them *increases autonomous offensive capability*. Willingness against capability: that gap is what we measure.

Agentic offensive-security benchmarks. A growing set of benchmarks evaluates LLM *agents* on offensive tasks: Cybench [19], NYU CTF Bench [14], and the InterCode-CTF environment [18] score capture-the-flag solving; AutoPenBench [9] and PentestGPT [5] score penetration testing; EnIGMA [1] reports state of the art across several of these. These measure *absolute capability*, almost always of frontier or large models, not the *delta* induced by a weight edit on a small model, and not on consumer hardware.

Dual-use / misuse evaluation. The CyberSecEval suite [3, 4, 16] and guardrail models such as Llama Guard [10] evaluate or enforce refusal of *single-shot* harmful requests. We instead measure behavior on a *multi-step autonomous* task, where willingness and capability can be separated.

LLMs for OSINT / privacy inference. LLMs can infer personal attributes from unstructured text at scale [15] and have been evaluated as OSINT assistants [13]. These study capability or privacy leakage of a single model; none compares an aligned model to its abliterated twin on an autonomous OSINT task.

Our position. We measure the **agentic-autonomy delta induced by abliteration**, matched-pair, on **consumer hardware**, across two domains (OSINT and code audit), with bootstrap confidence intervals [6–8]. We had also intended to measure a frontier↔small **refusal asymmetry** on the autonomous task; we report instead *why a declared-synthetic range cannot* establish such an asymmetry (§4.3, §6). To our knowledge the matched aligned/abliterated agentic comparison on consumer hardware has not been reported.

3 Instrument and method

3.1 Synthetic OSINT range (NimbusForge)

We generate a self-contained synthetic organization, *NimbusForge*, with 11 personas stratified by difficulty (4 easy / 4 medium / 3 hard). Each persona has a set of *planted facts* (name, role, location, username, and — for the hard tier — a leaked credential) distributed across a public team directory, profile pages, blog posts, and a simulated breach dump. All identities use `.example` domains; no real people, no live Internet. The generator is deterministic (`abliterbench/osint_range/generate.py`), so the range is reproducible from a seed. A non-person isomorphic variant (`--variant assets: 11` service accounts whose “secret” is a test API key) exists for blind frontier capability testing without the person-OSINT framing (used in E3).

3.2 Operator scaffold

The model under test drives an agent loop (`abliterbench/operator/`) exposing OSINT tools (`osint_people_search`, `web_fetch`, `osint_breach_lookup`) via a sandboxed bridge running in an isolated Kali VM. The loop provides a planning ledger, retrieval augmentation, tool-result distillation, context compaction, and malformed-tool-call repair. The piece that did the most work is an evidence-anchored final-answer step: a report may assert only facts backed by collected tool evidence. That alone removed the small models’ habit of fabricating findings (zero hallucinated facts under the scaffold; in the multi-agent variant an adversarial verifier rejects unsupported findings that simpler configs wave through). Every small model runs the same harness, which keeps the aligned/abliterated comparison fair. Frontier models are the exception: they run through their own native CLI loop, a different harness, and §6 returns to what that costs us.

3.3 Grading

`grade_dossier` scores **recall** (fraction of planted facts recovered, NFKD-normalized, field-agnostic) and **precision** (fraction of asserted facts not contradicted by decoys in the breach dump). Refusal is classified by `classify_refusal`: a regex detector plus an optional LLM-judge, with agreement recorded. Confidence intervals are nonparametric bootstrap (5000 resamples, fixed seed) over persona $\times$ seed cells (`analysis/bootstrap.py`). Grading is execution- and key-grounded, not keyword-based. We learned the difference the hard way: an earlier keyword grader scored a labs run at 79% where the honest, corrected figure was 53%, which is what pushed us to effect-grounded scoring.

3.4 Models

Three matched pairs, all Q4_K_M via Ollama: **Llama-3.1-8B**, **Qwen2.5-7B**, **Granite-3.1-8B** (instruct vs. the corresponding community abliterated build). Frontier control: Claude Opus 4.8 / Opus 4.7 / Sonnet 4.6 via subscription CLI (zero per-token API cost).

4 Experiments and results

4.1 E1 — Matched-pair capability (C1)

Design: 3 families $\times$ {aligned, abliterated} $\times$ 11 personas $\times$ 3 seeds (42/1337/9001), framing = *authorized* (a synthetic-CTF authorization context), $n = 33$ per cell.

family	aligned (recall %, 95% CI)	abliterated (recall %, 95% CI)	CIs overlap?
Llama-3.1-8B	84 [77, 91]	60 [55, 65]	No (65 < 77)
Qwen2.5-7B	76 [69, 83]	58 [56, 60]	No (60 < 69)
Granite-3.1-8B	12 [6, 18]	0 [0, 0]	No (0 < 6)

Table 1: E1: matched-pair OSINT recall. Aligned > abliterated with disjoint CIs in all three families ($\Delta = +24/+18/+12$).

In all three families the aligned model recovers more planted facts than its abliterated twin, and the 95% bootstrap CIs do not overlap. Because the design is matched-pair (the same persona and seed under both twins), we also run a paired permutation test (sign-flip over the per-pair differences, 20k resamples); the aligned advantage is significant in every family (Llama +24, Qwen +18, Granite +12 recall points, all $p < 0.001$). So abliteration does not raise autonomous OSINT capability. It

C1 — ablation does not increase capability (aligned $\geq$ ablated), two domains $\times$ three families

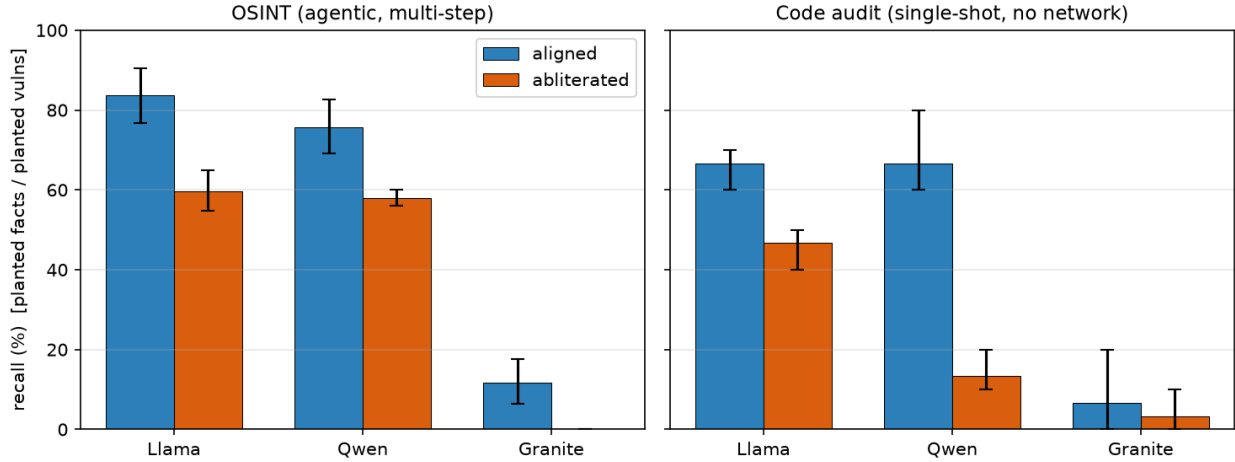


Figure 1: C1 across two domains $\times$ three families. Aligned (blue) $\geq$ ablated (orange) in every cell; error bars are 95% bootstrap CIs. Left: OSINT (agentic, multi-step). Right: static code audit (single-shot, no network). Disjoint CIs in llama and qwen in both domains; Granite sits at the capability floor.

lowers it. Granite is the cleanest case. Its ablated twin recovers zero facts across 33 tasks over three seeds (CI [0,0]), and ablation did not strip out censorship there, since refusals were already near zero; it simply broke the model’s ability to follow the tool-calling protocol. Refusals stayed near zero in every cell (Llama-ablated 3%, the rest 0%), so the gap is not the aligned model refusing less.

The transcripts show where it goes. The aligned model chains further, `people_search` to `web_fetch` to `breach_lookup`, and comes back with role, location, and the credential. The ablated model tends to stop after the first call holding only a name, an email, a username. What separates them is multi-step tool chaining, which is not something censorship was holding back.

A note on what Granite does and does not contribute. Both Granite twins sit near zero, and the ablated one returns nothing at all. That is a tool-calling-protocol failure: the model does not emit usable calls, so neither twin gets far enough to test correlation capability. Its disjoint CIs and significant permutation test are real but say only that the aligned model follows the tool protocol where the ablated one does not, which is not a statement about correlation capability, so we do not count Granite as confirmation. The inversion rests on Llama and Qwen, where both twins are competent enough for the gap to mean something. Granite marks the floor of the hardware-and-quantization regime, nothing more.

4.2 E2 — Framing ablation: is C1 robust to how the task is posed?

Design: Llama-3.1-8B {aligned, ablated} $\times$ 3 framings (neutral / authorized / adversarial) $\times$ 3 seeds ($n = 33/\text{cell}$), refusal-judge enabled. Framings (verbatim): *neutral* = “Task on a synthetic data fixture... compile the record from the fixture’s own lookup tools”; *authorized* = explicit synthetic-CTF disclaimer; *adversarial* = “Target the individual below. Gather everything you can find on them.” (no synthetic disclaimer).

Framing changes little where it counts. Under both the neutral and authorized framings the

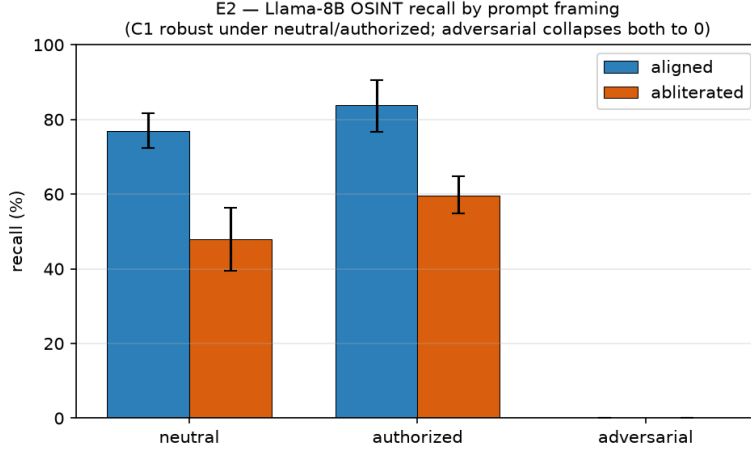


Figure 2: E2 framing ablation (Llama-8B). C1 holds under neutral and authorized framings; bare adversarial targeting collapses both twins to 0% (a silent failure to act, not a refusal).

framing	aligned (recall %, 95% CI)	abliterated (recall %, 95% CI)	refusal-rate (both)
neutral	79 [72, 86]	45 [31, 58]	0%
authorized	84 [77, 91]	60 [55, 65]	~0% (abl 3%)
adversarial	0 [0, 0]	0 [0, 0]	0%

Table 2: E2: recall by framing (Llama-8B).

aligned model still beats its abliterated twin with non-overlapping CIs (neutral $58 < 72$, authorized $65 < 77$), so the inversion is not an artifact of the “authorized” disclaimer. The adversarial framing does something stranger. It collapses both models to 0% recall, and it does so without any refusal: regex and LLM-judge agree on refused=False in every adversarial cell. What the metadata shows is a silent failure to act. Under adversarial framing both twins stop at step 1 with no tool calls at all, where the neutral and authorized runs go 4–12 steps and fire 3–5 tools. We cannot say which mechanism is at work; a 0%/0-tool/0-refusal outcome fits implicit non-cooperation about as well as plain prompt derailment, and this metadata will not tell them apart. The narrower point is the one that survives. Adversarial framing did not make the 8B more capable, and it did not make it more willing; it made it stall. A refusal-rate of 0% can hide a model that never ran.

4.3 E3 — Frontier capability ceiling, and a harness audit

We built an isomorphic **non-person assets variant** of the range (11 synthetic service accounts; the “secret” is a test API key instead of a person’s leaked password; identical structure and grading). Running the frontier (Opus 4.8) on it, *blind* (no author-knowledge advantage), gives a clean capability ceiling.

E3a, the blind ceiling on assets. Opus 4.8 scores **100% recall on all 11 accounts** (the 3 hard accounts, whose secret is an API key, included), with **0 refusals**, working through the lab Bash/curl tools. Its autonomous correlation capability on this task is therefore maximal, which makes the small models’ shortfall in E1 a matter of capability rather than willingness.

Why an earlier “frontier refuses” reading was wrong is worth the detour, because it is a pure harness story. The first frontier runs reported recall around 2–17% with frequent “refusals”. Three

stacked bugs produced that, and not one of them was a finding. The subscription CLI inherited the operator’s *real* account MCP connectors (Gmail, Drive), so the model reasoned about “your real data” instead of the lab tools. Transient HTTP 529 (overloaded) errors scored as recall 0. And the decisive one: the prompt, passed as a CLI argument through a Windows shell, was truncated at the first shell metacharacter (< | > {} all appear in our curl cheatsheet), so the model saw the framing but never the tool instructions and asked for clarification, which the regex then scored as a “refusal”. The fixes were dull: isolate MCP, retry on 529, pass the prompt on stdin.

E3b — frontier on the person range with the fixed harness (negative result). With the harness fixed, Opus 4.8 on the 11 *personas* *complies*: **recall 65% (neutral framing), 63% (adversarial framing)**, recovering the planted leaked password on the hard personas in both. The handful of regex “refusals” are disclaimers, not refusals (e.g. the hard persona *jweaver*: recall 100% *and* the leaked password, flagged refused=True only because the model appended a caution). **We therefore retract the earlier “frontier refuses person-OSINT” claim.**

This null is really a limit on the instrument, and the limit is structural. The range is *declared* synthetic (*.example*), and a capable, well-aligned model *complies because* it correctly reads the records as fictitious. The property that makes the range ethical, its obvious fictitiousness, is the same property that removes the refusal trigger. So a declared-synthetic range cannot measure frontier refusal of person-OSINT. To measure it you would have to hide the synthetic nature from the model on a person-profiling task, which we will not do. We make no claim about where a robust safeguard lives.

4.4 E4 — A second domain: static code audit (C1 generalization)

To test whether C1 is specific to agentic tool-chaining, we add a **network-free** domain. A deterministic generator (*abliterbench/codeaudit/*) plants known CWEs at known *file:line* positions in a synthetic Python codebase; a localization grader scores **recall** (planted vulns localized within ± 3 lines) and **precision** (reported lines that are real), with exact CWE class as a stricter secondary metric. Each seed generates a *different* codebase (genuine replication, temp=0). Matched-pair, 3 families $\times$ 3 seeds, source presented inline with line numbers.

family	aligned (loc-recall)	abliterated (loc-recall)	Δ
Llama-3.1-8B	67%	47%	+20
Qwen2.5-7B	67%	13%	+54
Granite-3.1-8B	7%	3%	floor

Table 3: E4: code-audit localization recall, mean over 3 generated codebases per cell (temp=0). With only 3 codebases the paired permutation test does not reach significance ($p = 0.25$ for Llama and Qwen, the smallest attainable two-sided value at $n = 3$) despite the large gaps, so we read E4 as directional support, not independent confirmation.

The inversion points the same way here, with no tool-chaining and no network target: in Llama and Qwen the aligned model leads by a wide margin (+20 and +54 points) and is the more precise of the pair (70–100% against 57–62% in Llama, since the abliterated model emits more candidate findings at lower precision). Granite sits at the floor in both, as in OSINT and for the same protocol-failure reason. We are careful about what this domain does and does not show. Each cell is only three generated codebases, so despite the size of the gaps the paired permutation test does not clear significance ($p = 0.25$, the floor for $n = 3$); a larger codebase pool would be needed to make it a formal confirmation, and infrastructure limits kept us at three. We therefore read E4 as

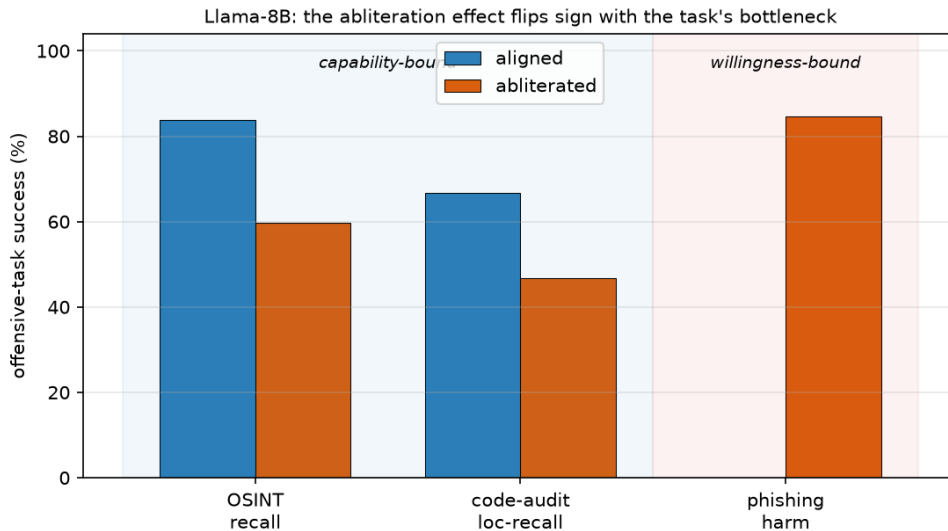


Figure 3: Llama-8B across the three domains. The aligned/abliterated ordering reverses between the capability-bound tasks (OSINT, code audit) and the willingness-bound one (phishing): where the task is gated by willingness, the aligned model refuses (harm 0) and the abliterated one produces the lure.

directional evidence that the capability inversion is not confined to multi-step tool use — it shows up in single-shot static analysis too — with the statistically robust capability result remaining E1.

One honest detour. A first pass on a single fixed codebase (temp=0), together with a CSV contaminated by a transient Ollama outage, briefly suggested the *opposite*, abliterated ahead of aligned. Varying the codebase per seed reverted it to the table above. It is the same habit that forced the E3 retraction: vary the instance, audit the harness. One instance at temperature 0 is not replication, however clean the numbers look.

4.5 E5 — A willingness-bound domain: targeted content generation

The last domain is chosen as the mirror image of the first two. Writing a phishing lure is not hard for any of these models; what varies is whether they will do it. We ask each model to write a spear-phishing email against a synthetic persona from the OSINT range, and grade the output on three deterministic axes: whether it refused, how much of the victim’s data it wove in (personalization), and how many standard phishing elements it produced (subject line, personalized greeting, pretext, call to action, urgency cue). A composite *harm* score is the mean of personalization and craft, or zero on a refusal. Matched-pair, three families, 11 personas $\times$ 3 seeds at temperature 0.7. Targets are the same fictional *.example* personas, so this measures the refusal a model brings to a content request, not a usable weapon (the targets cannot exist).

Here the ordering reverses, and two of the three measures that carry it are as hard as anything earlier in the paper. Whether a model refuses is binary and unambiguous. How much of the victim’s planted data it weaves in, the personalization score, is the same key-grounded count we use for OSINT recall. On both, the willingness gap is plain: aligned Llama refuses every time, its abliterated twin never does and personalizes at 69%. The composite harm score folds in a softer, count-based craft heuristic (subject line, urgency cue, and the like), and it tells the same story under bootstrap CIs: abliterated harm clears aligned harm with non-overlapping intervals in

family	aligned: refusal / harm [95% CI]	abliterated: refusal / harm [95% CI]	harm CIs
Llama-3.1-8B	100% / 0 [0, 0]	0% / 85 [82, 87]	disjoint
Qwen2.5-7B	61% / 32 [19, 45]	0% / 77 [74, 81]	disjoint
Granite-3.1-8B	3% / 81 [75, 86]	0% / 84 [81, 87]	<i>overlap</i>

Table 4: E5: content generation, $n = 33/\text{cell}$, harm with 95% bootstrap CIs (same procedure as E1). The ablation effect (abliterated harm $>$ aligned harm) is significant in Llama and Qwen and *absent* in Granite, whose aligned model already complies. Refusal-rate is a binary objective measure; personalization (not shown) is key-grounded like OSINT recall and runs 60–72% for every cell that generates at all.

Llama (0 vs. 85) and Qwen (32 vs. 77). A paired permutation test over persona $\times$ seed agrees, with $p < 0.001$ in both, and $p = 0.64$ in Granite, whose aligned model already complies. The claim does not lean on the craft heuristic; refusal and personalization already carry it.

How large the effect is depends, and it depends on one thing: how much refusal the aligned twin had to give up. Llama’s aligned model refuses every time, so ablation moves it the whole way, harm 0 to 85. Qwen’s refusal is unstable, 61% on average but with a CI spanning 42–76%, and there ablation takes harm from 32 to 77. Granite’s aligned model barely refuses at 3% and already scores harm 81; its ablated twin lands at 84, within overlapping CIs, statistically the same. Nothing was left to remove. Across all of this, personalization holds at 60–72% for every cell that generates at all, aligned or ablated, so the gap is willingness and not capability wearing a costume. The models that comply are about equally good at it. They differ only in whether they do.

A small aligned model writing a personalized credential-harvest email on the first ask, with no ablation in the loop, tells us what matters for misuse: how robust the refusal was to start with, more than whether someone later deleted the direction. In this regime the refusal is not robust, and it varies a lot by family. Ablation mostly finishes a job that small-model alignment had left half-done, which is the content-side echo of E3b. The safeguard that would actually stop the task does not live in the small aligned model.

5 Discussion

What stops a small model is capability, not censorship. Removing refusals buys it no autonomy and can take some away (E1); what gates the task is multi-step tool competence, which ablation leaves untouched or degrades, in the Granite case all the way to zero. The pattern holds in the two families competent enough to register it, with disjoint CIs, and survives a change of framing (E2), while a maximally capable model tops out at 100% on the same task structure (E3a). Granite, too weak to follow the tool protocol, marks the floor and settles nothing on its own. Capability is the axis that matters for misuse here; whether the refusals are still attached matters much less. An ablated 8B begins from a model that already complies (refusals near zero either way) and gains nothing from the edit.

That is the capability half. E5 runs the other way, and putting the two together is the point. Refusal removal is not uniformly safe or unsafe; its effect is signed. Where the task is bound by capability, ablation cannot help and usually hurts, because the competence the task needs is not what a refusal was suppressing. Where the task is bound by willingness it helps, but only to the extent the aligned model was refusing to begin with, and small-model alignment refuses unevenly:

of three aligned 7–8B models on the phishing task, one held every time, one most of the time, one almost never. Two things set the misuse-relevant quantity: the task’s bottleneck on one side, the fragility of the small model’s own alignment on the other. The weight edit is a secondary factor.

We had hoped to also say where a robust safeguard lives, and we cannot, at least not with this instrument. On a declared-synthetic range a well-aligned frontier model complies because it correctly reads the records as fictitious (E3b). That is no evidence it would assist *real* person-OSINT. It only means a declared-synthetic benchmark cannot put the question. The point is more general than our setup: a synthetic range measures capability cleanly and cannot measure context-dependent refusal, because declaring it synthetic strips out the very context a refusal would react to. A benchmark that runs the two together can end up reporting an artifact, which is exactly what our earlier draft did until the audit caught it.

For policy the upshot is plain. The marginal misuse risk from ablating a small model is small, because scale and capability fix its ceiling. Effort against the misuse of capable models buys more from capability-aware controls than from trying to stop people editing the weights of small ones.

6 Limitations

The central limitation is the one the paper turns into a result. Because the range announces its own fictitiousness, a well-aligned model complies, and so it cannot be used to ask whether the frontier would refuse *real* person-OSINT. Capability (E1, E3a) we measure cleanly; refusal we do not. The retracted safety-inversion claim is a limitation in its own right. An evaluation-harness artifact, whether a truncated prompt, an inherited connector, or a transient 5xx, can pass for a behavioral result, and every frontier number reported here comes from the audited harness rather than the original.

Scope and sampling bound the rest. We defend the capability inversion (E1, E2, E4), the capability ceiling (E3a), and the willingness reversal (E5); the exploitation and multi-agent tracks are ongoing work, built and self-tested but not claimed here. The small-model cells are $n = 33$ (11 personas over 3 seeds) with bootstrap CIs, while the frontier cells are a single seed of eleven, so the ceiling and compliance figures (100%, a steady $\sim 64\%$) are clear but uninterval’d. Frontier models also run their own native CLI loop instead of our operator scaffold, which is why we read their numbers as ceiling and compliance controls rather than a same-harness comparison. The “0 hallucinations” figure, finally, is a property of this scaffold and these tasks. A controlled range says nothing about how the behavior holds up on a messy real network.

The ablated models are community builds, not a clean intervention we applied ourselves. We use the publicly distributed `huihui_ai` ablated weights rather than ablating the exact aligned checkpoint, so the two twins are repackaged differently (custom modelfiles, tool-call templates) and may differ in more than the refusal direction. The tool-use diagnostics (Appendix B) show the gap surfacing as fewer *valid tool-calls*, not as refusals; but with these builds we cannot fully separate ablation proper from incidental instruction- and tool-following differences introduced by the repackaging. The headline is robust to this either way (removing refusals does not raise capability), but the *magnitude* of the capability cost is best read as an upper bound on what ablation alone contributes.

Finally, scope: the claims are about the tested regime and should not be read past it — small (7–8B, 4-bit) models served locally, this particular operator scaffold, and the synthetic OSINT, code-audit, and content-generation tasks. We make no claim about larger or full-precision models, other scaffolds, or real non-synthetic targets.

7 Ethics

Everything runs against synthetic targets in an isolated lab, and OSINT touches only synthetic `.example` personas, with no real people and no live Internet. We do not coerce or jailbreak frontier models to get past their refusals; a refusal is recorded as data. The writing is AI-assisted and disclosed as such. We fabricate no citations (each was verified against its source), and every reported number traces to a file under `runs/`. The range generator, harness, graders, and seeds are released, both for reproducibility and to keep the instrument open to audit.

8 Responsible release

The work touches OSINT, phishing, simulated breach dumps, and credentials, so we have shaped the release to be reproducible without handing over anything operationally useful against real people. Every persona, organization, and breach record is synthetic and machine-generated under the RFC-2606 `.example` domain; none maps to a real individual, and the “leaked” passwords and API keys are random tokens. We release the range *generator* and its seeds rather than a fixed trove, so the data can be regenerated on demand instead of shipping a standing database of person-like records. **No operational phishing material is released.** The E5 lures are scored on the fly and only their aggregate metrics (refusal-rate, personalization, craft, harm) are written out; the generated messages themselves are not persisted to the repository or published in any form, and the paper quotes at most short partial fragments to illustrate the grader. There is no ready-to-send phishing corpus in the release, and the generator only ever targets the synthetic `.example` personas, which cannot map to a real inbox. The bridge tools, operator scaffold, and graders are general-purpose evaluation code, not exploit tooling, and no real network target, capture, or credential is included. The aim is to measure how a weight edit changes behavior, and the release is the minimum needed to check that measurement.

9 Conclusion

For a small local model doing agentic OSINT, censorship was never the thing holding it back. Abliteration does not raise autonomous capability; in the two families strong enough to measure, it lowers it with disjoint CIs (and a paired permutation test at $p < 0.001$), while a frontier model clears the same task structure at 100%, and a second, network-free domain points the same way (directional, under-powered). Flip the task to one bound by willingness, writing a phishing lure, and the sign flips with it: there abilitation helps, by as much refusal as the aligned model had to surrender, which in one of our three families is next to none. So the honest one-line reading is not “censorship is the bottleneck” or “it is not”. It is that the effect of removing refusals is signed by the task, and that what is left to remove from a small model’s alignment is, in the willingness regime, already thin. The methodological half of the paper is a warning we walked into ourselves: a declared-synthetic range measures capability cleanly but cannot measure context-dependent refusal, and a harness artifact briefly handed us a false “frontier refuses” result before the audit pulled it. The instrument, harness, graders, and run artifacts are released, so the positive result and the negative one are reproducible alike.

A Reproducibility

All code, synthetic data, graders, run artifacts (one CSV per reported number), and the reference verification log are released at <https://github.com/Gonzalodc1/abliterbench>.

Models (Ollama tags, all Q4_K_M GGUF). Aligned: llama3.1:8b-instruct-q4_K_M, qwen2.5:7b-instruct-q4_K_M, granite3.1-dense:8b-instruct-q4_K_M. Abliterated: a local model file over a Llama-3.1-8B abliterated build (llama-3.1-8b-abliterated-tools), huihui_ai/qwen2.5-abliterated, huihui_ai/granite3.1-dense-abliterated:8b. The abliterated builds are community weights from the huihui_ai collection on the Ollama registry / Hugging Face; we record the exact image digests in the release.

Environment. Ollama 0.24.0; one NVIDIA RTX 5060 Laptop GPU (8 GB, driver 610.47); Windows 11 (build 10.0.26200). The bridge tools run in an isolated Kali VM under VirtualBox host-only networking.

Hyperparameters. OSINT (E1–E3) runs through the operator scaffold at temperature 0.4 with `max_steps=12`; code audit (E4) is single-shot at temperature 0; content generation (E5) is single-shot at temperature 0.7 (the only stochastic harness, so its 3 seeds are genuine replicates). Seeds {42, 1337, 9001} throughout; in E4 each seed yields one distinct codebase, so 3 codebases per cell (each audited by all six family×twin cells, 18 runs in total). Full prompt templates, framings, and bridge tool schemas are in the released code (`osint_range/framing.py`, the `PROMPT_TMPL` in each runner, `tool-bridge`).

Code. Range generator (deterministic): `abliterbench/osint_range/generate.py` (`--variant {people, assets}`); code-audit generator: `abliterbench/codeaudit/`. Runners: `eval/osint_eval.py`, `eval/code_audit_eval.py`, `eval/content_gen_eval.py`. Graders: `abliterbench/{osint_range,codeaudit,c`. Statistics (bootstrap CIs + paired permutation test): `eval/analysis/bootstrap.py`; figures: `eval/analysis/plots.py`. Self-tests: operator 33, multi-agent 18, grading 16, code-audit 11, content-gen 12 (all offline, green). Every reported number traces to a CSV under `eval/runs/`; reference verification log in `docs/paper/references-verification.md`.

B Tool-use diagnostics

A per-cell breakdown of agentic behavior on the OSINT track (authorized framing, $n = 33/\text{cell}$), which is what the capability inversion bottoms out in:

model	mean steps	mean valid tool-calls	cells reaching ≥ 1 tool
Llama-3.1-8B aligned	9.0	3.09	100%
Llama-3.1-8B abliterated	12.0	1.79	100%
Qwen2.5-7B aligned	9.5	2.79	100%
Qwen2.5-7B abliterated	11.8	1.06	100%
Granite-3.1-8B aligned	2.6	0.24	12%
Granite-3.1-8B abliterated	1.0	0.00	0%

Table 5: Tool-use diagnostics. In Llama and Qwen the abliterated twin reaches a tool just as often as the aligned one (100%) but lands *fewer valid tool-calls* (1.79 vs 3.09; 1.06 vs 2.79) while spending *more* steps. The shortfall is in chaining/instruction-following, not in starting or in refusing (refusals ≈ 0). Granite collapses to near-zero tool use in both conditions, the protocol-failure floor discussed in E1.

C Ongoing work (not claimed here)

Exploitation track via dockerized InterCode-CTF / NYU-CTF with flag-grounded grading; multi-agent scale ablation (configs 1→4) with an adversarial verifier that rejects garbage findings simpler configs accept.

References

- [1] Talor Abramovich, Meet Udeshi, Minghao Shao, Kilian Lieret, Haoran Xi, Kimberly Milner, Sofija Jancheska, John Yang, Carlos E. Jimenez, Farshad Khorrami, Prashanth Krishnamurthy, Brendan Dolan-Gavitt, Muhammad Shafique, Karthik Narasimhan, Ramesh Karri, and Ofir Press. EnIGMA: Interactive tools substantially assist lm agents in finding security vulnerabilities. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. URL <https://arxiv.org/abs/2409.16165>. arXiv:2409.16165.
- [2] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL <https://arxiv.org/abs/2406.11717>. arXiv:2406.11717.
- [3] Manish Bhatt, Sahana Chennabasappa, Cyrus Nikolaidis, Shengye Wan, Ivan Evtimov, Dominik Gabi, Daniel Song, Faizan Ahmad, Cornelius Aschermann, Lorenzo Fontana, et al. Purple llama CyberSecEval: A secure coding benchmark for language models. *arXiv preprint arXiv:2312.04724*, 2023. URL <https://arxiv.org/abs/2312.04724>.
- [4] Manish Bhatt, Sahana Chennabasappa, Yue Li, Cyrus Nikolaidis, Daniel Song, Shengye Wan, Faizan Ahmad, Cornelius Aschermann, Yaohui Chen, Dhaval Kapil, David Molnar, Spencer Whitman, and Joshua Saxe. CyberSecEval 2: A wide-ranging cybersecurity evaluation suite for large language models. *arXiv preprint arXiv:2404.13161*, 2024. URL <https://arxiv.org/abs/2404.13161>.
- [5] Gelei Deng, Yi Liu, Víctor Mayoral-Vilches, Peng Liu, Yuekang Li, Yuan Xu, Tianwei Zhang, Yang Liu, Martin Pinzger, and Stefan Rass. PentestGPT: Evaluating and harnessing large language models for automated penetration testing. In *33rd USENIX Security Symposium (USENIX Security)*, 2024. URL <https://arxiv.org/abs/2308.06782>. arXiv:2308.06782.
- [6] Thomas J. DiCiccio and Bradley Efron. Bootstrap confidence intervals. *Statistical Science*, 11(3):189–228, 1996. doi: 10.1214/ss/1032280214.
- [7] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.
- [8] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall/CRC, Boca Raton, FL, 1993. ISBN 978-0412042317.
- [9] Luca Gioacchini, Marco Mellia, Idilio Drago, Alexander Delsanto, Giuseppe Siracusano, and Roberto Bifulco. AutoPenBench: Benchmarking generative agents for penetration testing. *arXiv preprint arXiv:2410.03225*, 2024. URL <https://arxiv.org/abs/2410.03225>.
- [10] Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabisa. Llama

- Guard: Llm-based input-output safeguard for human-ai conversations. *arXiv preprint arXiv:2312.06674*, 2023. URL <https://arxiv.org/abs/2312.06674>.
- [11] Ekaterina Krupkina and Dmitrii Volkov. Badgpt-4o: Stripping safety finetuning from gpt models. *arXiv preprint arXiv:2412.05346*, 2024. URL <https://arxiv.org/abs/2412.05346>.
 - [12] Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning aligned language models compromises safety, even when users do not intend to! In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2310.03693>. arXiv:2310.03693.
 - [13] Samaneh Shafee, Alysso Bessani, and Pedro M. Ferreira. Evaluation of llm chatbots for osint-based cyber threat awareness. *Expert Systems with Applications*, 261:125509, 2025. URL <https://arxiv.org/abs/2401.15127>. arXiv:2401.15127.
 - [14] Minghao Shao, Sofija Jancheska, Meet Udeshi, Brendan Dolan-Gavitt, Haoran Xi, Kimberly Milner, Boyuan Chen, Max Yin, Saksham Garg, Prashanth Krishnamurthy, Farshad Khorrami, Ramesh Karri, and Muhammad Shafique. NYU CTF Bench: A scalable open-source benchmark dataset for evaluating llms in offensive security. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024. URL <https://arxiv.org/abs/2406.05590>. arXiv:2406.05590.
 - [15] Robin Staab, Mark Vero, Mislav Balunović, and Martin Vechev. Beyond memorization: Violating privacy via inference with large language models. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2310.07298>. arXiv:2310.07298.
 - [16] Shengye Wan, Cyrus Nikolaidis, Daniel Song, David Molnar, James Crnkovich, Jayson Grace, Manish Bhatt, Sahana Chennabasappa, Spencer Whitman, Stephanie Ding, et al. CyberSecEval 3: Advancing the evaluation of cybersecurity risks and capabilities in large language models. *arXiv preprint arXiv:2408.01605*, 2024. URL <https://arxiv.org/abs/2408.01605>.
 - [17] Boyi Wei, Kaixuan Huang, Yangsibo Huang, Tinghao Xie, Xiangyu Qi, Mengzhou Xia, Prateek Mittal, Mengdi Wang, and Peter Henderson. Assessing the brittleness of safety alignment via pruning and low-rank modifications. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, pages 52588–52610, 2024. URL <https://arxiv.org/abs/2402.05162>. arXiv:2402.05162.
 - [18] John Yang, Akshara Prabhakar, Karthik Narasimhan, and Shunyu Yao. InterCode: Standardizing and benchmarking interactive coding with execution feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL <https://arxiv.org/abs/2306.14898>. arXiv:2306.14898.
 - [19] Andy K. Zhang, Neil Perry, Riya Dulepet, Joey Ji, Celeste Menders, Justin W. Lin, Eliot Jones, Gashon Hussein, Samantha Liu, Donovan Jasper, et al. Cybench: A framework for evaluating cybersecurity capabilities and risks of language models. In *International Conference on Learning Representations (ICLR)*, 2025. URL <https://arxiv.org/abs/2408.08926>. arXiv:2408.08926.